



**INSTITUTE OF ENGINEERING & MANAGEMENT
SALT LAKE, KOLKATA**

LAB MANUAL

Year : 2023 - 2027

Course Name : Computer Networks Laboratory

Course Code : PCCCS692

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Branch : AIML

Computer Networks Laboratory (PCCCS692)

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Session : 2025-2026

Experiment No : 1

Title: Exploring Basic Networking Commands

Objective:

This lab introduces students to essential networking commands used for diagnosing, monitoring, and troubleshooting networks. By completing this assignment, students will develop hands-on experience with commands in Windows, Linux, or macOS.

Identifying Network Configuration

1. **Objective:** Understand the network configuration of your system.
2. **Steps**
 - I. Run the following commands to view network configuration details:
 - **Windows:** `ipconfig`
 - **Linux/macOS:** `ifconfig` or `ip addr`

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\LENOVO> ipconfig

Windows IP Configuration

Unknown adapter CloudflareWARP:

    Connection-specific DNS Suffix . : 
    IPv6 Address. . . . . : 2606:4708:110:8cc6:99c4:78d2:ed54:741d
    Link-local IPv6 Address . . . . . : fe80::f585:8d1a:175a:e8e6%38
    IPv4 Address. . . . . : 172.16.0.2
    Subnet Mask . . . . . : 255.255.255.255
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix . : 
    IPv4 Address. . . . . : 192.168.5.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix . : 
    IPv4 Address. . . . . : 192.168.113.1
    Subnet Mask . . . . . : 255.255.255.0
```

- II. Note the following:
 - IP Address **172.16.0.2**
 - Subnet Mask **255.255.255.225**
 - Default Gateway **fe80::f585:8d1a:175a:e8e6%38**

3. Questions

- I. What is your system's IP address?
An IP address, or Internet Protocol address, is a unique identifier assigned to a device connected to a network. It's like a postal address, allowing data to be sent and received between devices on a network. There are two main types of IP addresses: IPv4 and IPv6.
- II. What is the role of the default gateway in your network?
In a network, the default gateway serves as an access point or a router that sends data from one network to another. Think of it as the middleman between your local network and the internet. When devices on your network need to communicate with devices on another network (such as accessing a website), they send the data packets to the default gateway. The gateway then forwards the packets to their destination outside the local network.

Testing Network Connectivity

1. **Objective:** Use the `ping` command to test connectivity with other devices.
2. **Steps:**
 - I. Ping the following:
 1. Your default gateway.

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
PS C:\Users\LENOVO> ping 172.16.0.2

Pinging 172.16.0.2 with 32 bytes of data:
Reply from 172.16.0.2: bytes=32 time<1ms TTL=128
Reply from 172.16.0.2: bytes=32 time<1ms TTL=128
Reply from 172.16.0.2: bytes=32 time<1ms TTL=128
Reply from 172.16.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\LENOVO>
```

2.A public server (e.g., 8.8.8.8).

```
PS C:\Users\LENOVO> ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=80ms TTL=118
Reply from 8.8.8.8: bytes=32 time=76ms TTL=118
Reply from 8.8.8.8: bytes=32 time=61ms TTL=118
Reply from 8.8.8.8: bytes=32 time=60ms TTL=118

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 60ms, Maximum = 80ms, Average = 69ms
PS C:\Users\LENOVO> |
```

3.A domain name (e.g., www.google.com).

```
PS C:\Users\LENOVO> ping www.google.com

Pinging www.google.com [2404:6800:4002:807::2004] with 32 bytes of data:
Reply from 2404:6800:4002:807::2004: time=118ms
Reply from 2404:6800:4002:807::2004: time=112ms
Reply from 2404:6800:4002:807::2004: time=120ms
Reply from 2404:6800:4002:807::2004: time=118ms

Ping statistics for 2404:6800:4002:807::2004:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 112ms, Maximum = 120ms, Average = 117ms
PS C:\Users\LENOVO> |
```

- II. Record the round-trip time (RTT) for each ping.

3. Questions

- I. Was the ping to each target successful?
Yes, the ping was successful to each target but it may fail under various circumstances.

- II. If a ping failed, what might be the reason?

Following are the reasons for a ping to fail :

- **Network Connectivity Issues:** If there is no physical or wireless connection between your device and the target device, the ping will fail.
- **Firewall or Security Settings:** Firewalls or security settings on either the source or target device may block ping requests, causing the ping to fail.
- **Target Device is Unreachable:** The target device might be turned off, disconnected from the network, or experiencing its own network issues.

- **Incorrect IP Address or Hostname:** If the IP address or hostname you are pinging is incorrect or does not exist, the ping will not succeed.
- **Network Congestion:** High traffic on the network can result in packet loss, causing ping requests to time out and fail.

Tracing Routes

1. **Objective:** Use the traceroute command to map the path to a destination.

2. **Steps:**

I. Run:

- **Windows:** `tracert www.google.com`
- **Linux/macOS:** `traceroute www.google.com`

II. Observe the hops the packets take to reach the destination.

3. **Questions**

I. How many hops did it take to reach www.google.com?

It took a total of 30 (Eighteen) hops to reach www.google.com

II. Did any hops time out? If so, what could cause this?

No, 0 hops timed out. But in some cases there might be time-out.

When performing a traceroute, hops may time out for several reasons:

1. **Firewall or Security Settings:** Firewalls on intermediate routers or the target device may block traceroute packets, causing timeouts.
2. **Network Congestion:** Heavy traffic on the network can result in packet delays or losses, leading to timeouts.
3. **Unresponsive Routers:** Some routers are configured not to respond to traceroute requests, causing the trace to show timeouts at those hops.
4. **Rate Limiting:** Network devices may limit the rate of responses to traceroute packets to prevent overloading, resulting in occasional timeouts.
5. **Device Issues:** The router or network device may be experiencing issues or high CPU usage, causing it to drop traceroute packets.

```
PS C:\Users\LENOVO> arp -a

Interface: 192.168.5.1 --- 0x4
Internet Address      Physical Address      Type
192.168.5.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static

Interface: 192.168.31.92 --- 0x7
Internet Address      Physical Address      Type
192.168.31.1          8c-83-94-86-4d-20    dynamic
192.168.31.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
```

I. Identify:

- MAC addresses of devices in the cache.
- Corresponding IP addresses.

2. Questions

I. What is the purpose of the ARP cache?

The ARP (Address Resolution Protocol) cache serves a crucial role in network communication. Here's what it does:

1. **Maps IP Addresses to MAC Addresses:** The ARP cache stores the mappings between IP addresses (used for network-level routing) and MAC addresses (used for data link layer communication). This mapping allows devices to communicate within a local network.
2. **Improves Network Efficiency:** By keeping a record of the IP-to-MAC address mappings, the ARP cache eliminates the need to repeatedly broadcast ARP requests to resolve addresses, thus reducing network traffic and latency.
3. **Enhances Speed:** Cached entries mean quicker access to devices on the network, improving the speed of data transmission since the device does not have to wait for an ARP request/response cycle every time it communicates.

II. How can outdated ARP entries affect network communication?

Outdated ARP entries can cause several issues in network communication, including:

1. **Misrouted Packets:** If an ARP cache contains outdated mappings, packets might be sent to the wrong MAC address. This misdirection can lead to failed communication attempts.
2. **Communication Failures:** Devices expecting to communicate with a specific MAC address may be unable to establish a connection if the address has changed but the ARP cache hasn't been updated.
3. **Increased Network Traffic:** Devices might repeatedly attempt to communicate with the wrong MAC address, generating unnecessary network traffic and potentially leading to congestion.

Experiment No :2

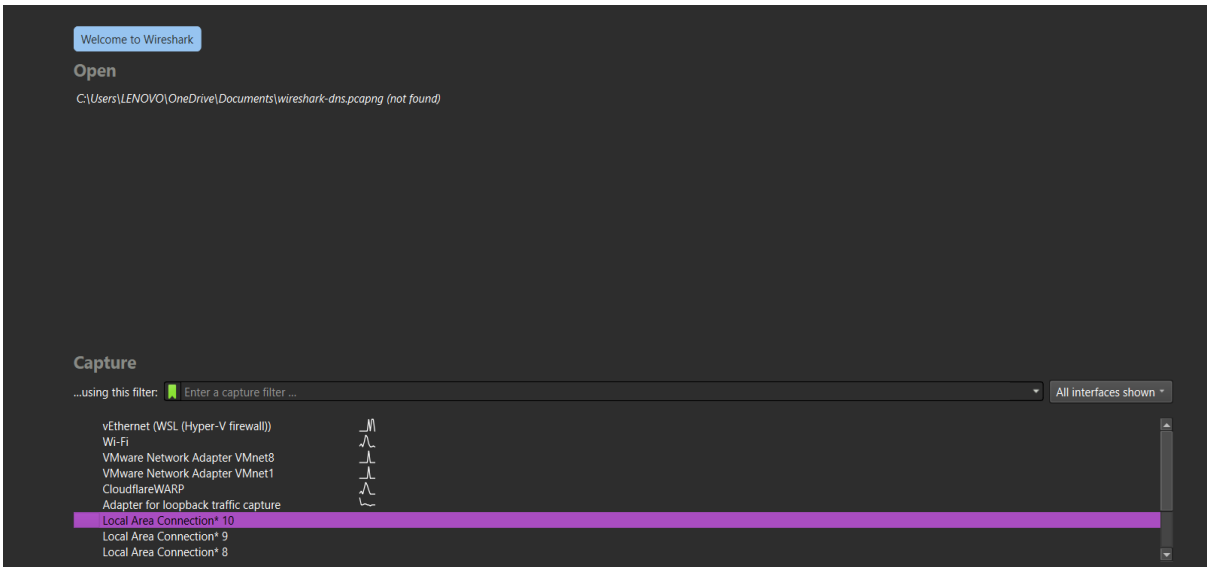
Title: Packet Capture and Analysis Using Wireshark

Objective:

This lab introduces students to network packet analysis using Wireshark. By completing the assignment, students will learn how to capture, filter, and analyse network traffic effectively.

Getting Started with Wireshark

- 1. **Objective:** Familiarize students with the Wireshark interface and basic functionality.
- 2. **Steps:**
 - a. Launch Wireshark and identify the available network interfaces.



- b. Start a packet capture on the primary interface (e.g., Wi-Fi or Ethernet).

No.	Time	Source	Destination	Protocol	Length	Info
15	2.327611	192.168.31.1	224.0.0.1	IGMPv3	50	Membership Query, general
16	2.591687	192.168.31.92	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
17	2.835234	192.168.31.92	162.159.198.2	UDP	125	62049 → 8443 Len=83
18	2.887175	162.159.198.2	192.168.31.92	UDP	88	8443 → 62049 Len=46
19	2.894584	162.159.198.2	192.168.31.92	UDP	125	8443 → 62049 Len=83
20	2.895038	192.168.31.92	162.159.198.2	UDP	88	62049 → 8443 Len=46
21	2.896220	192.168.31.92	162.159.198.2	UDP	125	62049 → 8443 Len=83
22	2.950296	162.159.198.2	192.168.31.92	UDP	88	8443 → 62049 Len=46
23	3.580849	192.168.31.92	224.0.0.22	IGMPv3	62	Membership Report / Join group 239.255.255.250 for any source
24	4.367336	162.159.198.2	192.168.31.92	UDP	145	8443 → 62049 Len=103
25	4.367749	192.168.31.92	162.159.198.2	UDP	88	62049 → 8443 Len=46
26	4.368540	192.168.31.92	162.159.198.2	UDP	145	62049 → 8443 Len=103
27	4.432983	162.159.198.2	192.168.31.92	UDP	88	8443 → 62049 Len=46
28	4.811731	192.168.31.92	162.159.198.2	UDP	84	62049 → 8443 Len=42
29	4.868432	162.159.198.2	192.168.31.92	UDP	88	8443 → 62049 Len=46
30	4.886405	162.159.198.2	192.168.31.92	UDP	125	8443 → 62049 Len=83
31	4.886756	192.168.31.92	162.159.198.2	UDP	88	62049 → 8443 Len=46
32	4.887648	192.168.31.92	162.159.198.2	UDP	125	62049 → 8443 Len=83

- 3. Browse a website (e.g., www.example.com) during the capture.
- 4. Stop the capture and save it as *capture_part1.pcap*.
- 5. **Questions**
 - a. Which network interface did you use, and why?
Wifi was used as it the available interface in my system connecting to the internet.

b. How many packets were captured in total?

```

▶ Frame 1: Packet, 145 bytes on wire (1160 bits), 145 bytes captured (1160 bits) on interface \Device\NPF_
▶ Ethernet II, Src: Arcadyan_86:4d:20 (8c:83:94:86:4d:20), Dst: Intel_83:22:c3 (c8:8a:9a:83:22:c3)
▶ Internet Protocol Version 4, Src: 162.159.198.2, Dst: 192.168.31.92
▶ User Datagram Protocol, Src Port: 8443, Dst Port: 62049
▶ Data (103 bytes)

```

Applying Filters

1. **Objective:** Learn to apply display filters to narrow down relevant packets.
2. **Tasks:**
 - a. Use the capture from Part 1.
 - b. Apply the following filters and note the results:
 - http (Display HTTP packets)

http						
No.	Time	Source	Destination	Protocol	Length	Info
820	6.582105	192.168.31.92	49.44.176.179	HTTP	208	GET /connecttest.txt HTTP/1.1
823	6.591176	192.168.31.92	49.44.176.179	HTTP	208	GET /connecttest.txt HTTP/1.1
827	6.615854	49.44.176.179	192.168.31.92	HTTP	241	[TCP Retransmission] HTTP/1.1 200 OK (text/plain)
829	6.615854	49.44.176.179	192.168.31.92	HTTP	241	[TCP Out-Of-Order] HTTP/1.1 200 OK (text/plain)
2270	42.081443	192.168.31.92	49.44.176.179	HTTP	208	GET /connecttest.txt HTTP/1.1
2273	42.091477	192.168.31.92	49.44.176.179	HTTP	208	GET /connecttest.txt HTTP/1.1
2278	42.118874	49.44.176.179	192.168.31.92	HTTP	241	[TCP Retransmission] HTTP/1.1 200 OK (text/plain)
2283	42.123131	49.44.176.179	192.168.31.92	HTTP	241	HTTP/1.1 200 OK (text/plain)
4387	112.792243	192.168.31.92	49.44.176.146	HTTP	201	GET /connecttest.txt HTTP/1.1
4396	112.865044	192.168.31.92	49.44.176.146	HTTP	201	GET /connecttest.txt HTTP/1.1
4400	112.896953	49.44.176.146	192.168.31.92	HTTP	241	HTTP/1.1 200 OK (text/plain)
4402	112.897375	49.44.176.146	192.168.31.92	HTTP	241	HTTP/1.1 200 OK (text/plain)
4972	148.414864	192.168.31.92	49.44.176.179	HTTP	201	GET /connecttest.txt HTTP/1.1
4975	148.422607	192.168.31.92	49.44.176.179	HTTP	201	GET /connecttest.txt HTTP/1.1
4978	148.441082	49.44.176.179	192.168.31.92	HTTP	241	[TCP Out-Of-Order] HTTP/1.1 200 OK (text/plain)
4983	148.457034	49.44.176.179	192.168.31.92	HTTP	241	HTTP/1.1 200 OK (text/plain)

- dns (Display DNS packets)
- ip.addr == <your IP> (Display packets related to your IP address)

ip.addr == 172.16.0.2						
No.	Time	Source	Destination	Protocol	Length	Info

c. Identify the DNS query and response for www.example.com.

ns.qry.name=="www.example.com"						
Time	Source	Destination	Protocol	Length	Info	
3693	92.708184	192.168.31.92	192.168.31.1	DNS	75	Standard query 0xb69a HTTPS www.example.com
3694	92.712553	192.168.31.92	192.168.31.1	DNS	75	Standard query 0xb8cc A www.example.com
3695	92.746410	192.168.31.1	192.168.31.92	DNS	117	Standard query response 0xb69a HTTPS www.example.com H
3696	92.749375	192.168.31.1	192.168.31.92	DNS	107	Standard query response 0xb8cc A www.example.com A 104

3. Questions

- a. What is the IP address resolved for www.example.com?
192.168.31.92
- b. How many HTTP packets were captured?
A total of 4123 packets were captured.

Analysing Protocols

1. **Objective:** Dive deeper into protocol details and packet structure.
2. **Steps:**
 - a. Select a single HTTP GET request packet.

No.	Time	Source	Destination	Protocol	Length	Info
820	6.582105	192.168.31.92	49.44.176.179	HTTP	208	GET /connecttest.txt HTTP/1.1
823	6.591176	192.168.31.92	49.44.176.179	HTTP	208	GET /connecttest.txt HTTP/1.1
827	6.615854	49.44.176.179	192.168.31.92	HTTP	241	[TCP Retransmission] HTTP/1.1 200 OK (text/plain)
829	6.615854	49.44.176.179	192.168.31.92	HTTP	241	[TCP Out-Of-Order] HTTP/1.1 200 OK (text/plain)
2270	42.081443	192.168.31.92	49.44.176.179	HTTP	208	GET /connecttest.txt HTTP/1.1
2273	42.091477	192.168.31.92	49.44.176.179	HTTP	208	GET /connecttest.txt HTTP/1.1
2278	42.118874	49.44.176.179	192.168.31.92	HTTP	241	[TCP Retransmission] HTTP/1.1 200 OK (text/plain)
2283	42.123131	49.44.176.179	192.168.31.92	HTTP	241	HTTP/1.1 200 OK (text/plain)
4387	112.792243	192.168.31.92	49.44.176.146	HTTP	201	GET /connecttest.txt HTTP/1.1
4396	112.865044	192.168.31.92	49.44.176.146	HTTP	201	GET /connecttest.txt HTTP/1.1
4400	112.896953	49.44.176.146	192.168.31.92	HTTP	241	HTTP/1.1 200 OK (text/plain)
4402	112.897375	49.44.176.146	192.168.31.92	HTTP	241	HTTP/1.1 200 OK (text/plain)
4972	148.414864	192.168.31.92	49.44.176.179	HTTP	201	GET /connecttest.txt HTTP/1.1
4975	148.422607	192.168.31.92	49.44.176.179	HTTP	201	GET /connecttest.txt HTTP/1.1
4978	148.441082	49.44.176.179	192.168.31.92	HTTP	241	[TCP Out-Of-Order] HTTP/1.1 200 OK (text/plain)
4983	148.457034	49.44.176.179	192.168.31.92	HTTP	241	HTTP/1.1 200 OK (text/plain)

- b. Expand the protocol layers (Ethernet, IP, TCP, HTTP) in the packet details pane.
- c. Note the source IP, destination IP, and the requested URL.

3. Questions

- a. What is the source and destination IP of the HTTP packet?
Source – 192.168.31.92
Destination – 49.44.176.179
- b. What is the URL requested in the GET packet?
208 GET /connecttest.txt HTTP/1.1

Capturing Specific Traffic

1. **Objective:** Use capture filters to focus on specific traffic.
2. **Tasks:**
 - a. Restart Wireshark and apply the following capture filter: `port 53`(DNS traffic).
 - b. Initiate a new DNS query by visiting a new website (e.g., www.google.com).
 - c. Stop the capture and save it as `capture_part4.pcap`
3. **Questions**
 - a. What is the DNS query sent for www.google.com?
Standard query 0x9498 A play.google.com
 - b. What was the response from the DNS server?
Standard query response 0x0788 AAAA play.google.com AAAA 2404:6800:4009:81f::200e

Experiment No : 3

Title : Study of Different Network Topologies Using Cisco Packet Tracer

Introduction:

The topology of a network defines how various devices like routers, switches, and computers are connected and how data flows between them. Different topologies offer distinct benefits and drawbacks in terms of performance, reliability, and cost. Cisco Packet Tracer is a simulation tool used to model various network topologies and test their functionality without the need for physical hardware. This report studies five common network topologies and aims to analyze their performance and behavior within a simulated environment.

Objective :

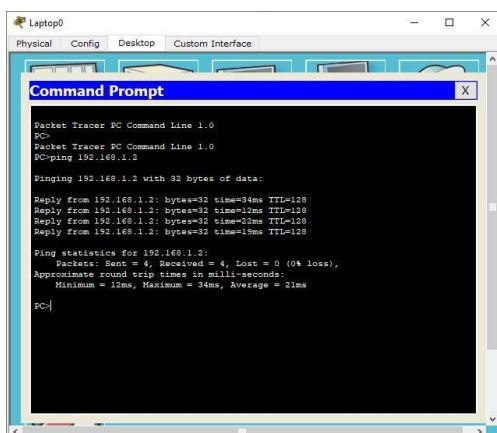
- To study and simulate various network topologies using Cisco Packet Tracer.
- To analyze and compare the advantages and disadvantages of different topologies.
- To evaluate the performance and behavior of each topology under different scenarios.

Material and Methods :

- **Cisco Packet Tracer:** A network simulation software used to design and analyze network topologies.
- **Network Devices:** Routers, switches, PCs, and other devices used to form the network.
- **Network Topologies:** The following topologies were studied:
 - Bus Topology
 - Star Topology
 - Ring Topology
 - Mesh Topology
 - Hybrid Topology

For each topology, a simulated network was created in Cisco Packet Tracer. The following steps were followed:

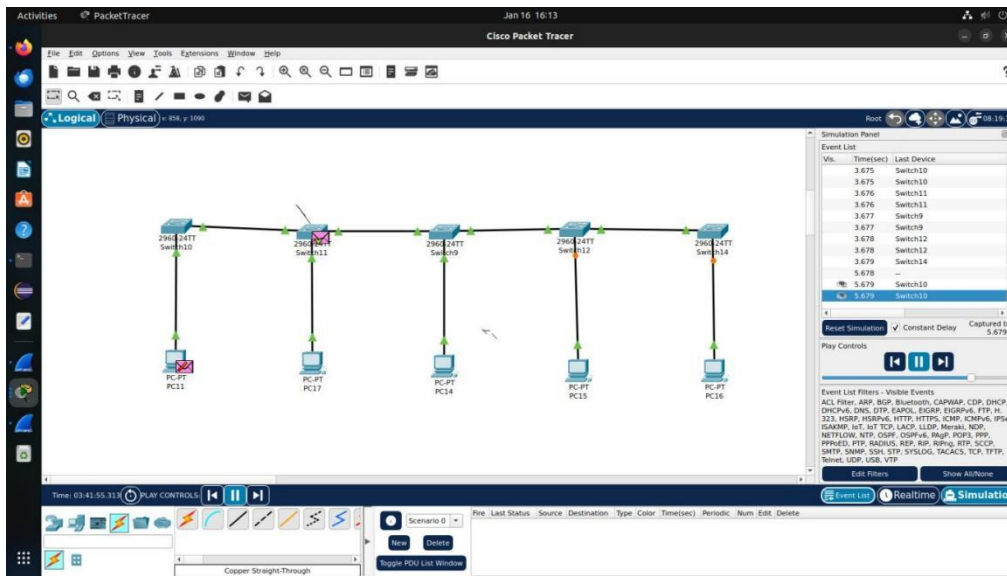
- Devices were connected to create the network according to the chosen topology.
- The appropriate IP configurations were applied to each device.
- Connectivity and performance were tested using ping commands and data transfers.
- Troubleshooting and failure scenarios were simulated to observe how the network behaved under fault conditions.



Theory :

Bus Topology:

- All devices are connected to a single central cable (bus).
- Data sent by any device is received by all others.
- Advantages: Easy to set up, cost-effective.
- Disadvantages: A failure in the central bus can bring down the entire network.

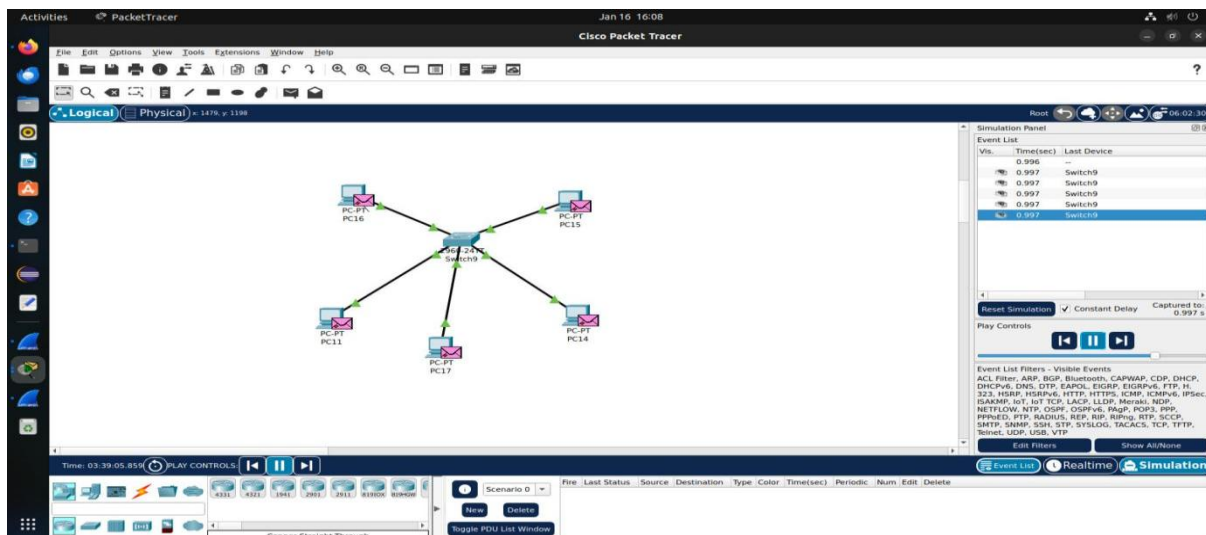


Observation :

When testing bus topology, we observed that network performance was good with minimal devices. However, as more devices were added, the network speed decreased due to congestion on the shared bus.

Star Topology:

- All devices are connected to a central device, usually a switch or hub.
- Advantages: Easy to manage and troubleshoot, failure in one device doesn't affect others.
- Disadvantages: High dependency on the central hub; if the hub fails, the entire network is affected.

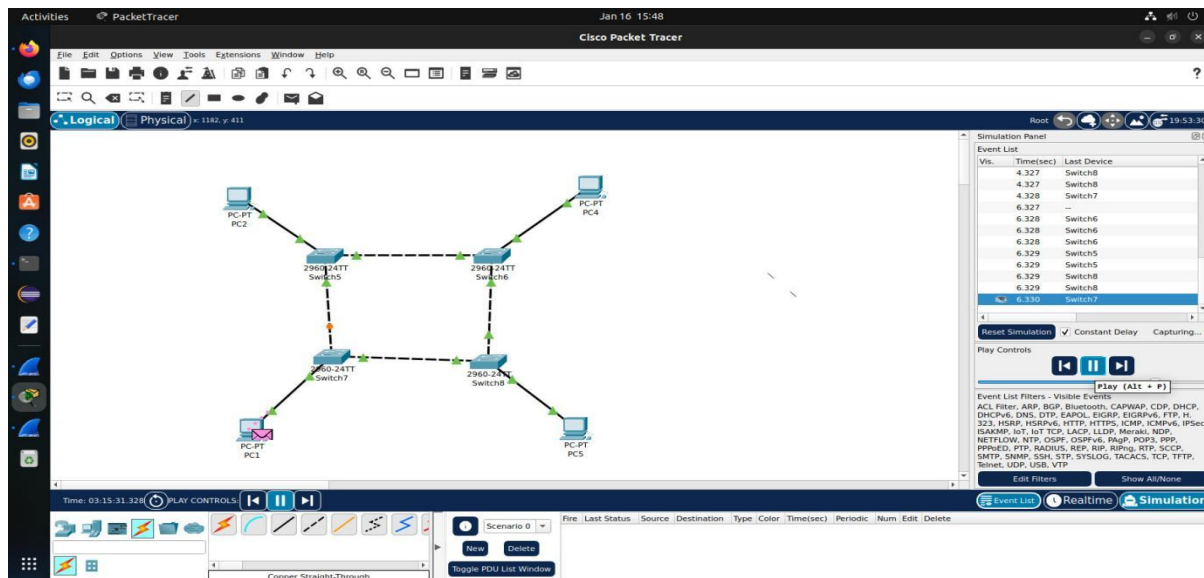


Observation :

The star topology was very stable, and adding/removing devices did not impact the overall network performance as long as the central hub/switch remained functional. However, a failure in the hub led to a total network shutdown.

Ring Topology:

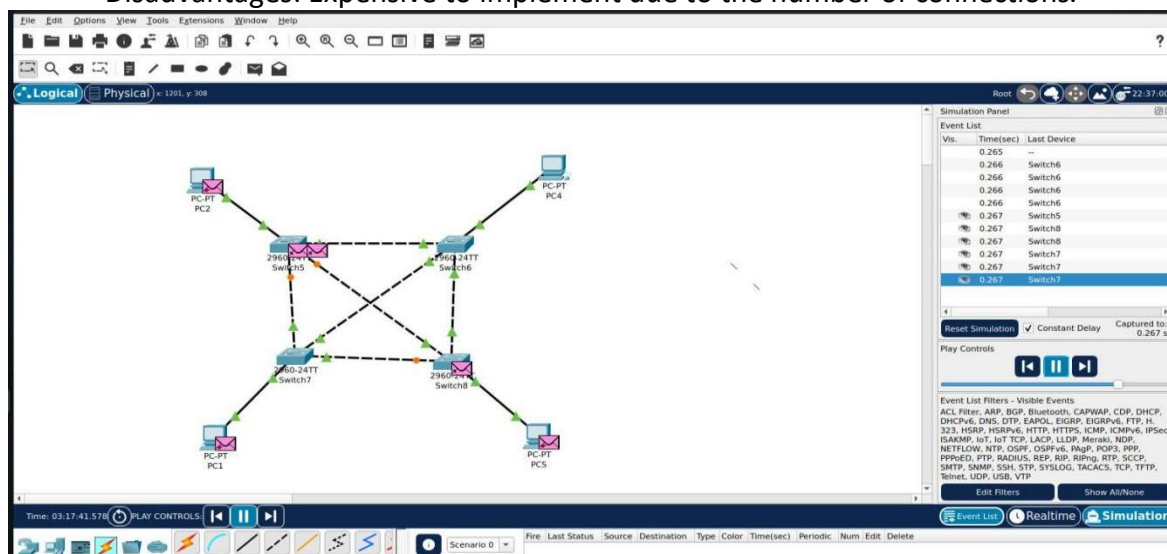
- Devices are connected in a circular fashion.
- Data travels in one direction (or two in a dual-ring topology).
- Advantages: Minimal collision; better performance with fewer devices.
- Disadvantages: A failure in one device or connection can disrupt the entire network.

**Observation :**

In the ring topology, data was transmitted smoothly without collisions. A failure in one of the connections, however, brought the entire network to a halt. In a dual-ring setup, the network remained functional even with a failed connection.

Mesh Topology:

- Every device is connected to every other device.
- Advantages: High redundancy, highly reliable.
- Disadvantages: Expensive to implement due to the number of connections.

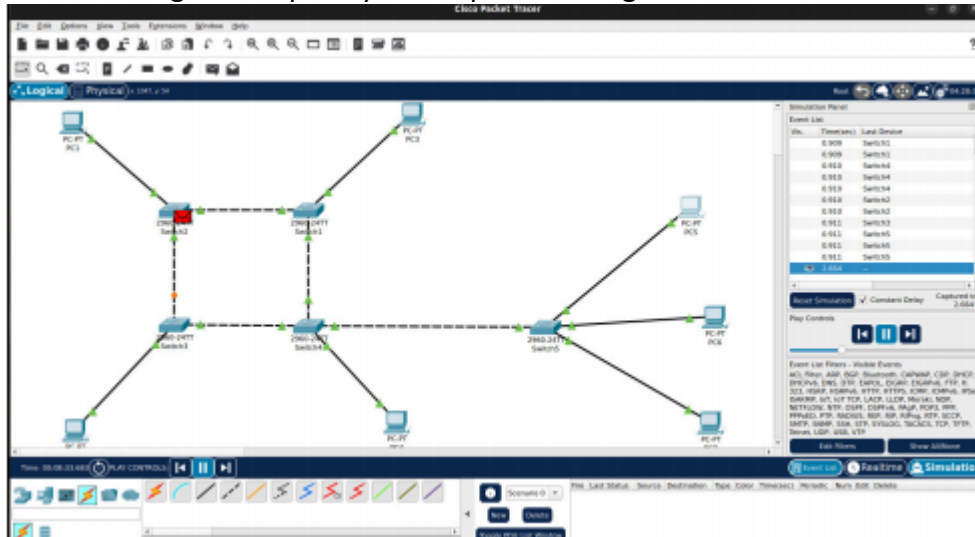


Observation :

Mesh topology proved to be the most resilient, with multiple paths for data to travel. Even if one link failed, communication could still be maintained. However, it was also the most resource-intensive topology, requiring the most connections and devices.

Hybrid Topology:

- A combination of two or more different topologies.
- Advantages: Flexible and scalable, can provide the benefits of different topologies.
- Disadvantages: Complexity in setup and management.

**Observation :**

Combining the best features of star and bus topologies provided a flexible and scalable solution. However, it was more complex to configure and manage.

Experiment No : 4

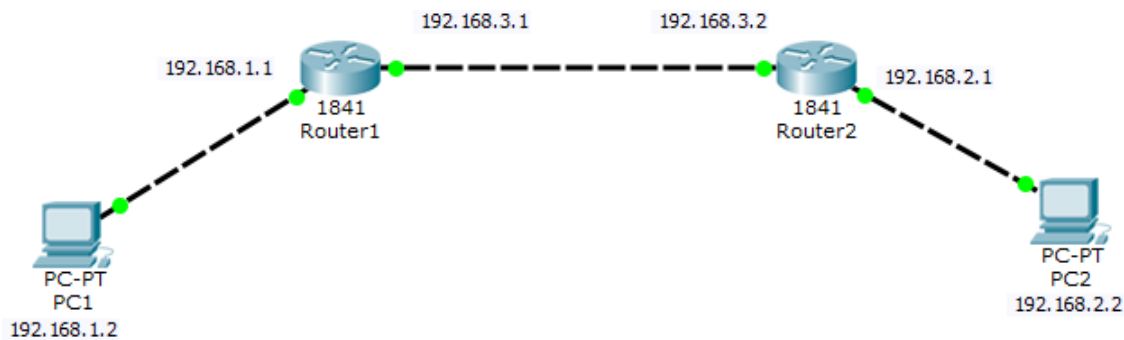
Title: Static Routing

Objective :

Implement static routing in Cisco Packet Tracer.

Routing is simply a process of choosing route for delivering data to its destination. All hosts who can generate a routing table can do a routing. Routing process is needed when we are going to deliver packets of data to a network that isn't directly connected with the sender.

For easier configuration process, let's send the data between 2 PCs across just 2 routers like the following network topology:



Step 1 : Place the routers.

First thing first, we need to place the routers on the workspace. We can pick one of them from *Network Devices > Routers > 4331* in bottom panel.

Any type of routers will do, 4331 is just the nearest to our cursor's current position.



Retrack the same step for another router, or copy paste the existing one.

Step 2 : Place the PCs.

PC can be picked from *End Devices > End Devices > PC*. Place them on the workspace.



Step 3 : Connect all hosts physically, and accordingly.

We can pick one of the physical media from *Connections > Connections*.

We are going to use ethernet connection, so we pick between **Straight-Through** or **Cross-Over** copper cable (choose **Straight** if unsure).



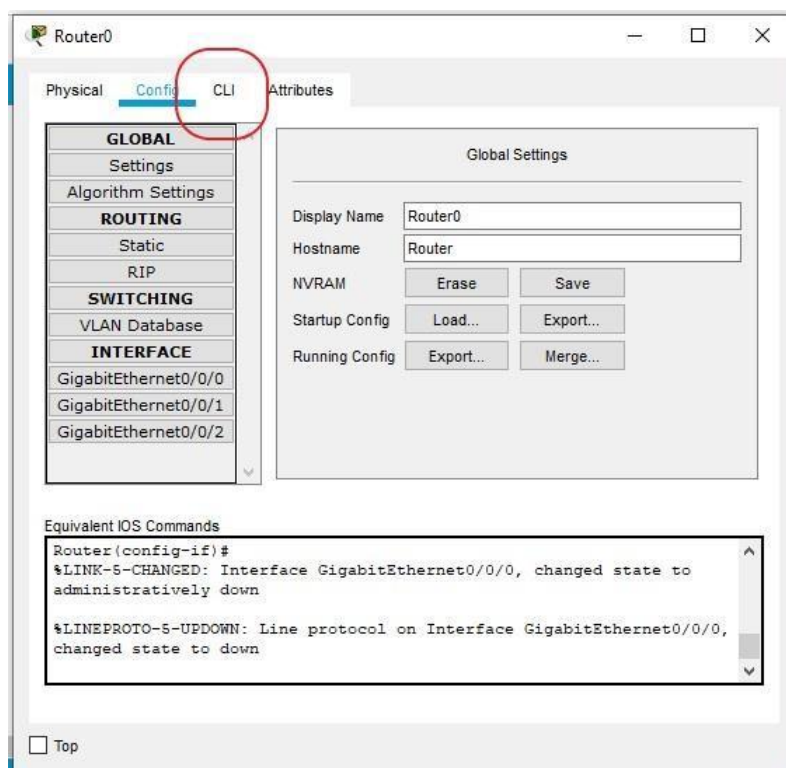
Next, click on one of the hosts, and choose one of the available ethernet ports. Then click on the another host while we see the cable being dragged.



Repeat the similar steps for connecting a PC and one of the routers.

Step 4 : Configure the routers.

Click on a router to bring up a pop-up window. Then open CLI tab.



In Router0 CLI, paste the following script:

```
en
con
f t
int FastEthernet0/0
ip address 192.168.1.1 255.255.255.0
no
shutdown
exit
int FastEthernet0/1
ip address 192.168.3.1 255.255.255.248
no
shutdown
exit
ip route 192.168.2.2 255.255.255.0 192.168.3.2
exit
```

And the following into Router1:

```
en
con
f t
int FastEthernet0/0
ip address 192.168.3.2 255.255.255.248
no
shutdown
exit
int FastEthernet0/1
ip address 192.168.2.1 255.255.255.0
no
shutdown
exit
ip route 192.168.1.0 255.255.255.0 192.168.3.1
exit
```

Type “no” for answering the following question beforehand if popping up.

Would you like to enter the initial configuration dialog? [yes/no]:

Step 5 : Configure the PCs.

There is Desktop tab in each PC's pop-up window. We are going to use IP Configuration and Command Prompt application here.



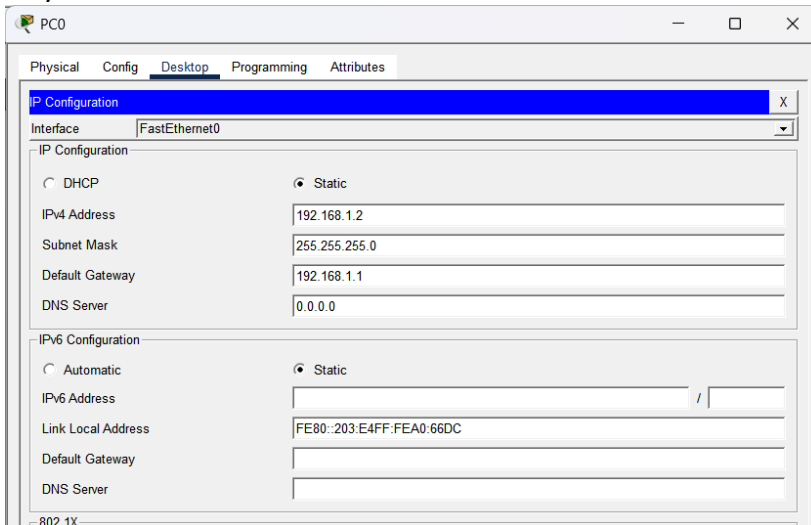
Choose IP Configuration to do IP configurations, and insert the following information.

For PC0.

IP Address : 192.168.1.2

Netmask : 255.255.255.0

Gateway : 192.168.1.1

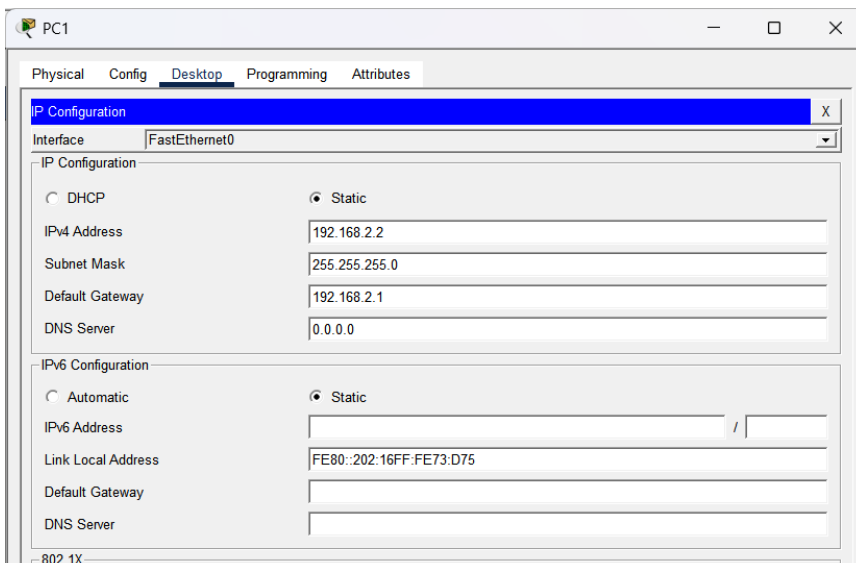


For PC1.

IP Address : 192.168.2.2

Netmask : 255.255.255.0

Gateway : 192.168.2.1



Step 6 : Ping test between the two PCs.

The image displays two screenshots of the Cisco Packet Tracer interface, showing a network topology and the results of ping tests between two PCs.

Top Screenshot: The network topology shows two routers, R4331 (Router0) and R4332 (Router1), connected to each other. PC-PT PC0 is connected to Router0, and PC-PT PC1 is connected to Router1. The Command Prompt window for PC0 shows the following output:

```
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>arp -a
Invalid Command.
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=6ms TTL=128
Reply from 192.168.1.2: bytes=32 time=6ms TTL=128
Reply from 192.168.1.2: bytes=32 time=6ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 6ms, Average = 3ms
C:\>
```

Bottom Screenshot: The network topology is the same as the top screenshot. The Command Prompt window for PC1 shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

Experiment No : 5

Title: Dynamic Routing Using Cisco Packet Tracer: RIP

Dynamic routing is a networking technique that provides optimal data routing. Unlike static routing, dynamic routing enables routers to select paths according to real-time logical network layout changes.

Advantages of Dynamic Routing

- Allows the exchange of routing information whenever the network experiences a change in topology.
- Since the routes do not have to be configured manually, there is less administrative overhead.
- Less error-prone than static routing.
- Allows scalability since there is less administrative overhead involved.

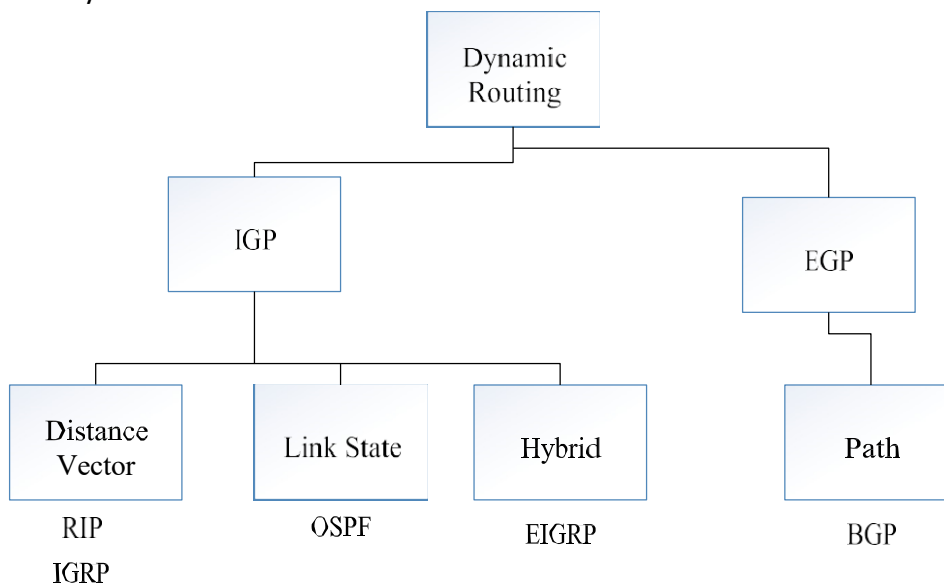


Fig: Classification of Dynamic Routing.

- An Autonomous System is a group of networks that is governed and controlled by a single administrative entity. For example, a network created by a single company, organization, corporation, or ISP is a single AS.
- An AS number is a unique identity of the AS on the Internet. If you want to connect your AS to the Internet, you must obtain an AS number. Internet Assigned Numbers Authority (IANA) has the worldwide right to assign AS numbers. It delegates that right to the organizations that assign public IP addresses. For example, in Asia, the Asia Pacific Network Information Centre (APNIC) assigns both IP addresses and AS numbers.
- An AS can contain multiple networks. To connect these networks, the administrator can use a routing protocol. The routing protocol the administrator uses to connect the networks within the autonomous system is known as an interior gateway routing protocol. Since all networks in an autonomous system belong to the same administrative entity, the administrator can configure any IGP protocol to connect them.
- RIPv1, IGRP, OSPF, EIGRP, RIPv2, and IS-IS are some examples of interior gateway routing protocols.
- An EGP protocol provides connectivity between different autonomous systems. Since different autonomous systems belong to different administrative entities, administrators cannot use routing protocols of their choices to connect them. They have to use a uniform routing protocol. A uniform routing protocol that connects different autonomous systems is known as an exterior gateway routing protocol.
- Nowadays, BGP is the only used exterior routing protocol. BGP connects all public autonomous systems on the Internet.

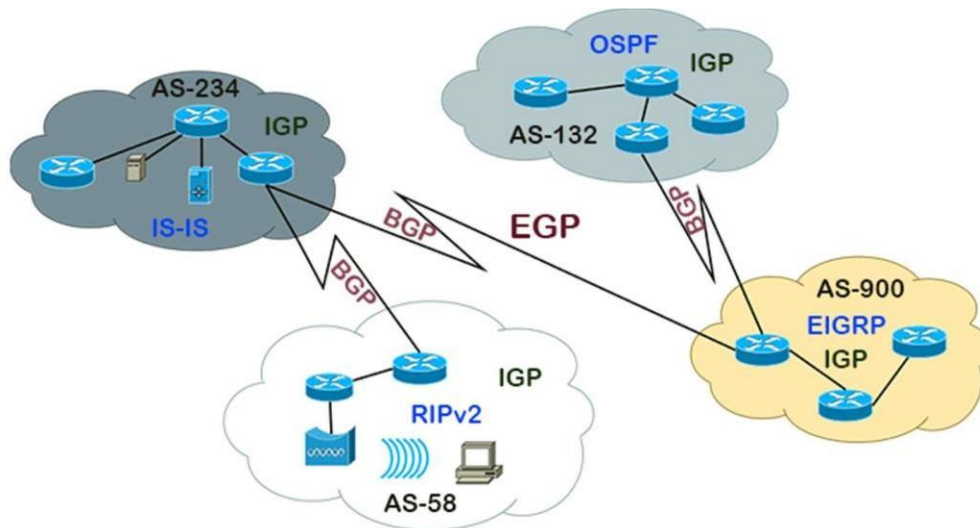


Fig: Schematic of Routing Protocols.

Distance Vector Routing

As the name implies, distance vector routing uses the parameters: distance and direction to calculate the best possible path to forward the packet from source to destination. Distance is calculated by counting the numbers of routers in between the source and destination. Every router counts its adjacent router as a next-hop. The minimum number of hop count between the source router and the destination router makes the best possible path for routing.

Distance vector routing uses the Bellman-Ford Algorithm or Ford-Fulkerson algorithms to choose the best path. In distance vector routing, routing advertisements are received from the directly connected or the neighbour router and update its routing information periodically. The update process continues router to router till the destination is reached. It shares entire routing table information to its immediate neighbour router.

Examples of distance vector routing protocols are RIP v1, RIPv2, and IGRP.

Link State Routing

Unlike distance-vector, link-state routing does not rely on the neighbour router for routing information. Instead, it maps the overall state of the entire network that includes the topology, bandwidth, traffic congestion in a particular link, etc.

In link-state routing, each router retrieves information about itself, its directly connected link, and the state of that link. This information is passed from router to router, without changing the entire routing information of the adjacent router. Each router makes only a copy of the information received from its neighbour. Thus, every router has identical information and can independently calculate its best path for forwarding packets from source to destination.

Link State routing uses Dijkstra's Algorithm. It is also referred to as the Shortest Path First (SPF) Algorithm.

Although link-state is more complex than distance-vector, however, it is more reliable and minimizes the shortcomings of distance vector routing. The basic functionality of link-state routing are:

- Every router establishes an adjacency relationship with its neighbours.
- Link state advertisements (LSAs) are sent by each router to its neighbour.
- Each router stores a copy of all the LSAs as a database. These databases in all routers are identical.
- Each router maintains a complete topological database, also called the link-state database.

- Each router calculates the shortest path to each network using Dijkstra's Algorithm and enters this information into the route table.

Examples of link-state routing are OSPF, IS-IS, etc.

Hybrid Routing

It is a combination of both distance vector and link-state routing. Basically, hybrid routing is considered to be the advanced distance-vector, however, it also contains some of the features of the link-state routing protocol.

An example of a hybrid routing protocol is EIGRP. Enhanced Interior Gateway Routing Protocol has the features of both the distance vector and the link-state routing. It does not send Link state advertisement, but it sends traditional distance vector routing information to the neighbour router. It synchronizes with the routing updates of the neighbour router. After that, it sends a specific link-state update to its neighbour.

10.a

Objective : To understand the concept and operation of Routing Information Protocol (RIP)

Requirements

- Windows pc – 2 Nos
- CISCO Packet Tracer Software (Student Version)
- 8 port switch – 2 No
- Router – 2 Nos
- Cat-5 LAN cable

Procedure

- Open the CISCO Packet tracer software
- Drag and drop 5 pcs using End Device Icons on the left corner
- Select 8 port switch from switch icon list in the left bottom corner
- Select Routers and Give the IP address for serial ports of router and apply clock rate as per the table.
- Make the connections using Straight through Ethernet cables
- Ping between PCs and observe the transfer of data packets in real and simulation mode.

Theory

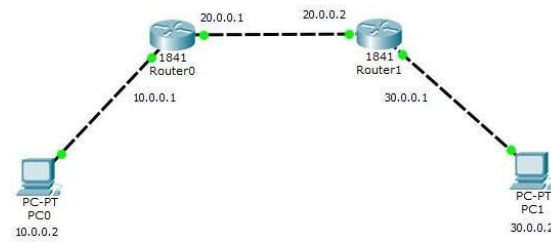
RIP (Routing Information Protocol) is one of the oldest distance vector routing protocols. It is usually used on small networks because it is very simple to configure and maintain, but lacks some advanced features of routing protocols like OSPF or EIGRP. Two versions of the protocol exists: version 1 and version 2. Both versions use hop count as a metric and have the administrative distance of 120. RIP version 2 is capable of advertising subnet masks and uses multicast to send routing updates, while version 1 doesn't advertise subnet masks and uses broadcast for updates. Version 2 is backwards compatible with version 1. RIPv2 sends the entire routing table every 30 seconds, which can consume a lot of bandwidth. RIPv2 uses multicast address of 224.0.0.9 to send routing updates, supports authentication and triggered updates (updates that are sent when a change in the network occurs).

Downsides of RIP:

- RIP sends the entire routing table every 30 seconds, which can consume a lot of network bandwidth.
- It lacks some more advanced of features of the newer routing protocols like OSPF or EIGRP and is not widely used in modern networks. For example, RIP doesn't support route summarization.

Network Topology Diagram for RIP 1.

1. Build the network topology.



2. Configure IP addresses on the PCs and the routers.

Router 1:

```

enable
configure
terminal
hostname
Router0

interface FastEthernet0/0
ip address 10.0.0.1
255.255.255.0 no
shutdown
exit

interface Serial0/0/0
ip address 20.0.0.1 255.255.255.252
clock rate
64000 no
shutdown
exit

exit
write memory
  
```

Router 2:

```

enable
configure
terminal
hostname
Router1

interface FastEthernet0/0
ip address 30.0.0.1
255.255.255.0 no
shutdown
exit

interface Serial0/0/0
ip address 20.0.0.2 255.255.255.252
no
shutdown
exit

exit
write memory
  
```

IP configuration on PCs

Click PC->Desktop->IP Configuration. On each PC assign these addresses:

PC1: IP address: 10.0.0.2; Subnet mask 255.0.0.0 Default Gateway 10.0.0.1

PC2: IP address: 30.0.0.2; Subnet mask 255.0.0.0 Default Gateway 30.0.0.1

3. Configure RIPv2 on the routers.

Router 1

```
enable
configure
terminal
router rip
version 2
network 10.0.0.0
network
20.0.0.0 no
auto-summary
exit
write memory
```

Router 2

```
enable
configure
terminal
router rip
version 2
network 30.0.0.0
network
20.0.0.0 no
auto-summary
exit
write memory
```

As you can see, to configure rip on each router, we enable RIP using router rip command then advertise the networks directly connected to the router interfaces using network command.

That's all for RIP configuration.

4. We'll now verify RIP configuration.

To verify that RIP is indeed advertising routes, we can use the show ip route command on R1.

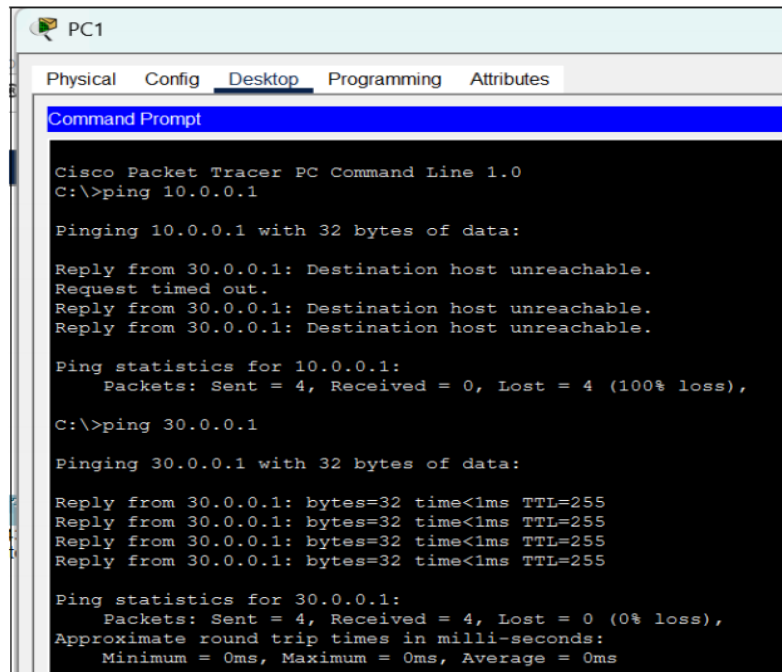
```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    20.0.0.0/8 is directly connected, FastEthernet0/1
R    30.0.0.0/8 [120/1] via 20.0.0.2, 00:00:13, FastEthernet0/1
_
```

You can see that R1 has learned about the 30.0.0/8 network. The letter R indicates that the route was learned using RIP. Note the administrative distance of 120 and the metric of 1 in the [120/1] part.

To specifically display routes learnt through RIP use show ip route rip command on the router.
Now lets Ping PC2 from PC1 to further confirm that connectivity is really established between the two subnets.



The screenshot shows a PC1 window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 30.0.0.1: Destination host unreachable.
Request timed out.
Reply from 30.0.0.1: Destination host unreachable.
Reply from 30.0.0.1: Destination host unreachable.

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 30.0.0.1

Pinging 30.0.0.1 with 32 bytes of data:

Reply from 30.0.0.1: bytes=32 time<1ms TTL=255
Reply from 30.0.0.1: bytes=32 time<1ms TTL=255
Reply from 30.0.0.1: bytes=32 time<1ms TTL=255
Reply from 30.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 30.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Experiment No : 6

Title: Domain Name System (DNS) Configuration and Analysis

Objective:

- Understand how DNS works and its role in network communication.
- Set up a local DNS server in a simulated environment.
- Configure and test DNS query resolution.
- Use Wireshark to capture and analyze DNS traffic.

Lab Requirements:

- A computer with Linux (Ubuntu/CentOS) or Windows (WSL or Virtual Machine recommended).
- BIND9 (for Linux) or Simple DNS Plus (for Windows) installed.
- Wireshark for packet capture and analysis.
- Internet access (optional, if testing external domains).

Part 1: Setting Up a Local DNS Server Step

1: Install a DNS Server

For Linux (Ubuntu/Debian):

1. Open the terminal and install **BIND9**:
`sudo apt update`
`sudo apt install bind9 -y`
2. Start and enable the DNS service:
`sudo systemctl start bind9`
`sudo systemctl enable bind9`

For Windows (Using Simple DNS Plus or Windows DNS Server):

1. Download and install **Simple DNS Plus** or enable **Windows DNS Server** via Server Manager.
2. Configure the primary DNS zone in the **DNS Manager**.

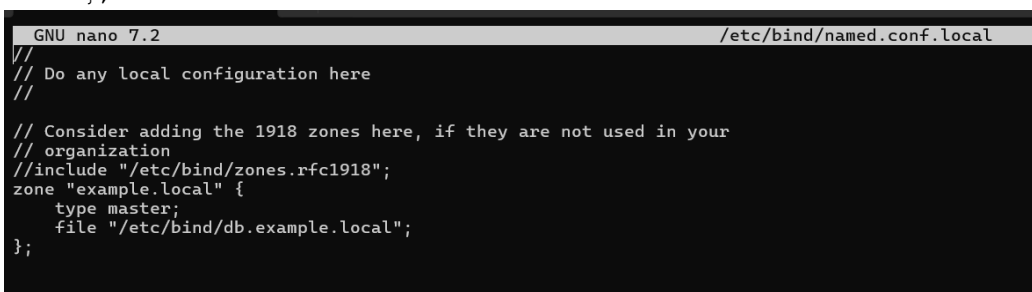
Step 2: Configure a Local DNS Zone (For Internal Domains)

1. Open the **BIND configuration file**:

```
sudo nano /etc/bind/named.conf.local
```

2. Add the following zone configuration for a **local domain (example.local)**:

```
zone "example.local" {type
master;
file "/etc/bind/db.example.local";
};
```



```
GNU nano 7.2 /etc/bind/named.conf.local
//
// Do any local configuration here
//
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";
zone "example.local" {
    type master;
    file "/etc/bind/db.example.local";
};
```

3. Create a **zone file** for the domain:

```
sudo nano /etc/bind/db.example.local
```

4. Add the following **DNS records**:

```
$TTL 604800
@IN SOA ns1.example.local. admin.example.local. (
2                ; Serial
604800           ; Refresh
86400            ; Retry
2419200         ; Expire
604800 )         ; Negative Cache TTL
; Name Servers
@ IN NS ns1.example.local.
; A Records
ns1 IN A 192.168.1.1
www IN A 192.168.1.2
```

```
GNU nano 7.2 /etc/bind/db.example.local
$TTL 604800
@ IN SOA ns1.example.local. admin.example.local. (
2                ; Serial
604800           ; Refresh
86400            ; Retry
2419200         ; Expire
604800 )         ; Negative Cache TTL
; Name Servers
@ IN NS ns1.example.local.
; A Records
ns1 IN A 127.0.0.1
www IN A 192.168.1.2
```

5. Restart the DNS service:

```
sudo systemctl restart bind9
```

6. Verify the DNS server status:

```
sudo systemctl status bind9
```

```
@LAPTOP-G00MI528:~$ sudo systemctl start bind9
@LAPTOP-G00MI528:~$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset>
   Active: active (running) since Fri 2026-03-13 00:12:34 UTC; 7min ago
     Docs: man:named(8)
    Main PID: 1438 (named)
      Status: "running"
    Tasks: 62 (limit: 4597)
   Memory: 38.0M ()
    CGroup: /system.slice/named.service
            └─1438 /usr/sbin/named -f -u bind
```

Part 2: DNS Query Resolution Testing

Step 1: Configure a Client to Use the Local DNS Server

1. Edit the **resolv.conf** file on the client system:

```
sudo nano /etc/resolv.conf
```

2. Add the following:

```
nameserver 192.168.1.1 domain
example.local
```

Step 2: Test DNS Resolution Using nslookup and dig

Using nslookup: `nslookup www.example.local`

Expected Output:

Server: 192.168.1.1

Address: 192.168.1.1#53

Name: www.example.local Address: 192.168.1.2

```
@LAPTOP-G00MI528:~$ nslookup www.example.local
Server:      10.255.255.254
Address:     10.255.255.254#53

** server can't find www.example.local: NXDOMAIN

@LAPTOP-G00MI528:~$ dig www.example.local
```

Using dig: `dig www.example.local`

Expected Output: `www.example.local.`

`604800 IN A 192.168.1.2`

```
@LAPTOP-G00MI528:~$ dig www.example.local
; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> www.example.local
; global options: +cmd
; Got answer:
; WARNING: .local is reserved for Multicast DNS
; You are currently testing what happens when an mDNS query is leaked to DNS
; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 994
; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; QUESTION SECTION:
;www.example.local.      IN      A

; Query time: 10 msec
; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
; WHEN: Fri Mar 13 00:22:13 UTC 2026
; MSG SIZE rcvd: 46
```

Part 3: DNS Packet Analysis with Wireshark Step

1: Capture DNS Traffic

1. Open **Wireshark** and select the **network interface** connected to the DNS server.
2. Start **packet capture** and apply the filter: `dns`
3. Open a **new terminal** and run:
`nslookup www.example.local`
4. Observe the **DNS query and response** packets in Wireshark.

2: Analyze DNS Query Resolution

Look for:

1. **DNS Request (Standard Query A)** – Sent by the client to the DNS server.
2. **DNS Response (Standard Query Response)** – Reply from the DNS server with the IP address.
1. **Query Type (A, AAAA, MX, NS, etc.)** – Identifies the requested record type.
2. **Query Time** – Measures DNS resolution time.

Experiment No :7

Title : Socket Programming

Objective :

To write a code in Java and C to implement Client-Server model where the server listens for incoming connection and client initiates the connection. Also to develop a Multithreaded server to connect and serve multiple clients.

Inputs :

Client.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <time.h>

int main() {
    int client_socket;
    struct sockaddr_in server_addr;
    char buffer[1024];
    time_t now;
    struct tm *timeinfo;
    char timestamp[80];

    client_socket = socket(AF_INET, SOCK_STREAM, 0);
    if (client_socket == -1) {
        perror("socket");
        exit(1);
    }

    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = inet_addr("127.0.0.1");
    server_addr.sin_port = htons(12345);

    if (connect(client_socket, (struct sockaddr*)&server_addr, sizeof(server_addr)) == -1)
    {
        perror("connect");
        close(client_socket);
        exit(1);
    }

    printf("Connected to the server\n");
    printf("Type your messages below (type 'exit' to quit):\n");

    while (1) {
        printf("Client: ");
        fgets(buffer, sizeof(buffer), stdin);
        buffer[strcspn(buffer, "\n")] = '\0'; // Remove the newline character

        time(&now);
```

```
    timeinfo = localtime(&now);
    strftime(timestamp, sizeof(timestamp), "%Y-%m-%d %H:%M:%S", timeinfo);

    snprintf(buffer, sizeof(buffer), "[%s] %s", timestamp, buffer);
    send(client_socket, buffer, strlen(buffer), 0);

    if (strcmp(buffer, "[%s] exit") == 0) {
        break;
    }

    int read_size = recv(client_socket, buffer, sizeof(buffer) - 1, 0);
    buffer[read_size] = '\0';
    printf("Server: %s\n", buffer);
}

close(client_socket);
printf("Connection closed\n");
return 0;
}
```

Server.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <pthread.h>
#include <time.h>

void *handle_client(void *arg);

int main() {
    int server_socket, client_socket;
    struct sockaddr_in server_addr, client_addr;
    socklen_t client_addr_size;
    pthread_t thread_id;

    server_socket = socket(AF_INET, SOCK_STREAM, 0);
    if (server_socket == -1) {
        perror("socket");
        exit(1);
    }

    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(12345);

    if (bind(server_socket, (struct sockaddr*)&server_addr, sizeof(server_addr)) == -1) {
        perror("bind");
        close(server_socket);
        exit(1);
    }
    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(12345);
```

```
if (bind(server_socket, (struct sockaddr*)&server_addr, sizeof(server_addr)) == -1) {
    perror("bind");
    close(server_socket);
    exit(1);
}

if (listen(server_socket, 5) == -1) {
    perror("listen");
    close(server_socket);
    exit(1);
}

printf("Server is listening on port 12345\n");

while (1) {
    client_addr_size = sizeof(client_addr);
    client_socket = accept(server_socket, (struct sockaddr*)&client_addr,
&client_addr_size);
    if (client_socket == -1) {
        perror("accept");
        continue;
    }
    printf("New client connected\n");

    if (pthread_create(&thread_id, NULL, handle_client, (void*)&client_socket) != 0) {
        perror("pthread_create");
        close(client_socket);
    }
    pthread_detach(thread_id);
}
close(server_socket);
return 0;
}

void *handle_client(void *arg) {
    int client_socket = *(int*)arg;
    char buffer[1024];

    int
read_size; time_t
now;
    struct tm *timeinfo;
    char timestamp[80];
    while ((read_size = recv(client_socket, buffer, sizeof(buffer) - 1, 0)) > 0) {
        buffer[read_size] = '\0';
        time(&now);
        timeinfo = localtime(&now);
        strftime(timestamp, sizeof(timestamp), "%Y-%m-%d %H:%M:%S", timeinfo);
        printf("[%s] Received message from client: %s\n", timestamp, buffer);

        if (strcmp(buffer, "exit") == 0) {
            break;
        }
    }
}
```

```
        snprintf(buffer, sizeof(buffer), "[%s] Message received: %s", timestamp, buffer);
        send(client_socket, buffer, strlen(buffer), 0);
    }
    close(client_socket);
    printf("Client disconnected\n");
    return NULL;
}
```

Outputs :

Client 1

```
@LAPTOP-G00MI528:~$ ./client
Connected to the server
Type messages (type 'exit' to quit)
Client: Zara
Server: [2026-03-15 08:56:11] Message received: [2026-03-15 08:56:11] Zara
Client: hello
Server: [2026-03-15 08:56:20] Message received: [2026-03-15 08:56:20] hello
Client: hi
Server: [2026-03-15 08:56:22] Message received: [2026-03-15 08:56:22] hi
Client: exit
Connection closed
```

Client 2

```
@LAPTOP-G00MI528:~$ ./client
Connected to the server
Type messages (type 'exit' to quit)
Client: new client
Server: [2026-03-15 08:57:00] Message received: [2026-03-15 08:57:00] new client
Client: bye
Server: [2026-03-15 08:57:05] Message received: [2026-03-15 08:57:05] bye
Client: why
Server: [2026-03-15 08:57:10] Message received: [2026-03-15 08:57:10] why
Client: exit
Connection closed
@LAPTOP-G00MI528:~$
```

Server:-

```
@LAPTOP-G00MI528:~$ ./server
Server is listening on port 12345...
New client connected
[2026-03-15 08:56:11] Received message: [2026-03-15 08:56:11] Zara
[2026-03-15 08:56:20] Received message: [2026-03-15 08:56:20] hello
[2026-03-15 08:56:22] Received message: [2026-03-15 08:56:22] hi
New client connected
[2026-03-15 08:57:00] Received message: [2026-03-15 08:57:00] new client
[2026-03-15 08:57:05] Received message: [2026-03-15 08:57:05] bye
[2026-03-15 08:57:10] Received message: [2026-03-15 08:57:10] why
[2026-03-15 08:57:14] Received message: [2026-03-15 08:57:14] exit
Client disconnected
[2026-03-15 08:57:48] Received message: [2026-03-15 08:57:48] exit
Client disconnected
```

Experiment No : 8

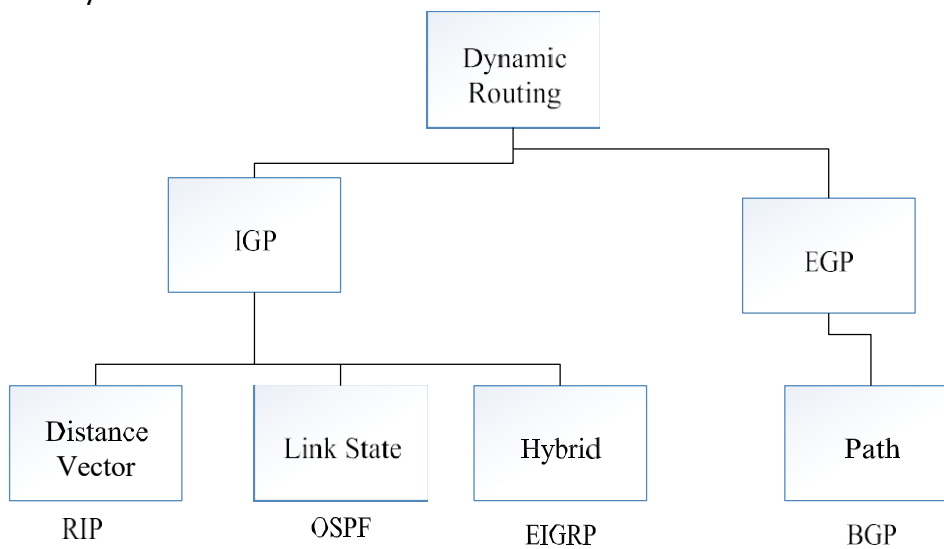
Title: Dynamic Routing Using Cisco Packet Tracer: OSPF, and BGP

OBJECTIVE:

Dynamic routing is a networking technique that provides optimal data routing. Unlike static routing, dynamic routing enables routers to select paths according to real-time logical network layout changes.

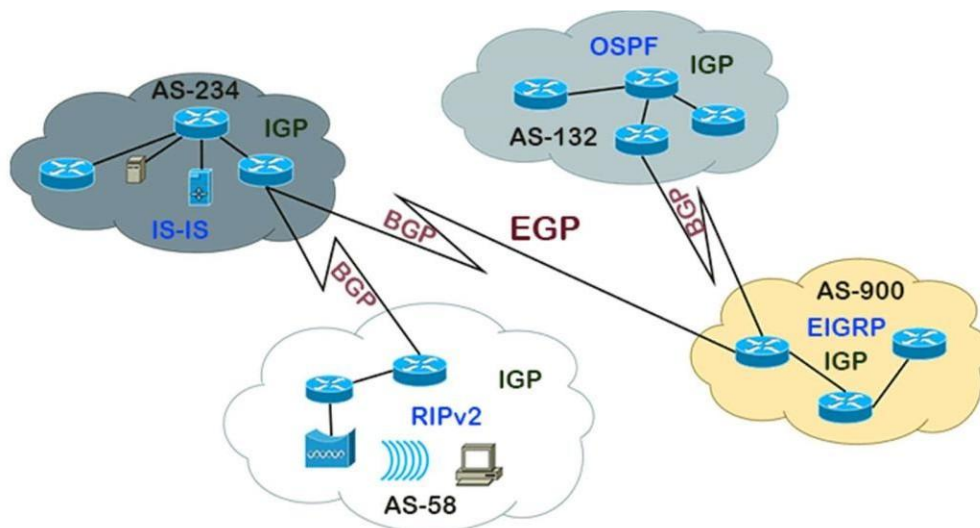
Advantages of Dynamic Routing

- Allows the exchange of routing information whenever the network experiences a change in topology.
- Since the routes do not have to be configured manually, there is less administrative overhead.
- Less error-prone than static routing.
- Allows scalability since there is less administrative overhead involved.



Classification of Dynamic routing

- An Autonomous System is a group of networks that is governed and controlled by a single administrative entity. For example, a network created by a single company, organization, corporation, or ISP is a single AS.
- An AS number is a unique identity of the AS on the Internet. If you want to connect your AS to the Internet, you must obtain an AS number. Internet Assigned Numbers Authority (IANA) has the worldwide right to assign AS numbers. It delegates that right to the organizations that assign public IP addresses. For example, in Asia, the Asia Pacific Network Information Centre (APNIC) assigns both IP addresses and AS numbers.
- An AS can contain multiple networks. To connect these networks, the administrator can use a routing protocol. The routing protocol the administrator uses to connect the networks within the autonomous system is known as an interior gateway routing protocol. Since all networks in an autonomous system belong to the same administrative entity, the administrator can configure any IGP protocol to connect them.
- RIPv1, IGRP, OSPF, EIGRP, RIPv2, and IS-IS are some examples of interior gateway routing protocols.
- An EGP protocol provides connectivity between different autonomous systems. Since different autonomous systems belong to different administrative entities, administrators cannot use routing protocols of their choices to connect them. They have to use a uniform routing protocol. A uniform routing protocol that connects different autonomous systems is known as an exterior gateway routing protocol.
- Nowadays, BGP is the only used exterior routing protocol. BGP connects all public autonomous systems on the Internet.



Schematic of routing protocols

10.b

Configuration of Open shortest Path First (OSPF) Algorithm : To construct multiple router networks and understand the operation of OSPF Protocol.

Requirements

- Windows pc – 3 Nos
- CISCO Packet Tracer Software (Student Version)
- 8 port switch – 3 No
- Router – 3 Nos
- Cat-5 LAN cable

Procedure

- Open the CISCO Packet tracer software.
- Drag and drop 5 pcs using End Device Icons on the left corner.
- Select 8 port switch from switch icon list in the left bottom corner.
- Select Routers and Give the IP address for serial ports of router and apply clock rate.
- Add HWIC -2T Peripheral to all routers, type CLI's for all routers.
- Make the connections using Straight through Ethernet cables.
- Ping between PCs and observe the transfer of data packets in real and simulation mode.

Theory

The OSPF routing protocol has largely replaced the older Routing Information Protocol (RIP) in corporate networks. Using OSPF, a router that learns of a change to a routing table (when it is reconfigured by network staff, for example) or detects a change in the network immediately multicasts the information to all other OSPF hosts in the network so they will all have the same routing table information. Unlike RIP, which requires routers to send the entire routing table to neighbour nodes every 30 seconds, OSPF sends only the part that has changed and only when a change has taken place. When routes change -- sometimes due to equipment failure -

- the time it takes OSPF routers to find a new path between endpoints with no loops (which is called "open") and that minimizes the length of the path is called the convergence time.

PC0	PC1	PC2
IP Address : 192.168.1.2 Gate way : 192.168.1.1	IP Address: 192.168.2.2 Gate way : 192.168.2.1	IP Address: 192.168.3.2 Gate way : 192.168.3.1

Router 0	Router 1	Router 2
fa 0/0 IP Address: 192.168.1.1 Serial 0/0/0 : 10.0.0.1 @ 2000000 clock rate Serial 0/0/1 : -	fa 0/0 IP Address : 192.168.2.1 Serial 0/0/0 : 10.0.0.2 Serial 0/0/1 : - @ 2000000 clock rate	Fa 0/0 IP Address : 192.168.3.1 Serial 0/0/0 : 10.0.0.2 @ no clock rate Se 0/0/1 : 11.0.0.1

Commands to Configuring OSPF:

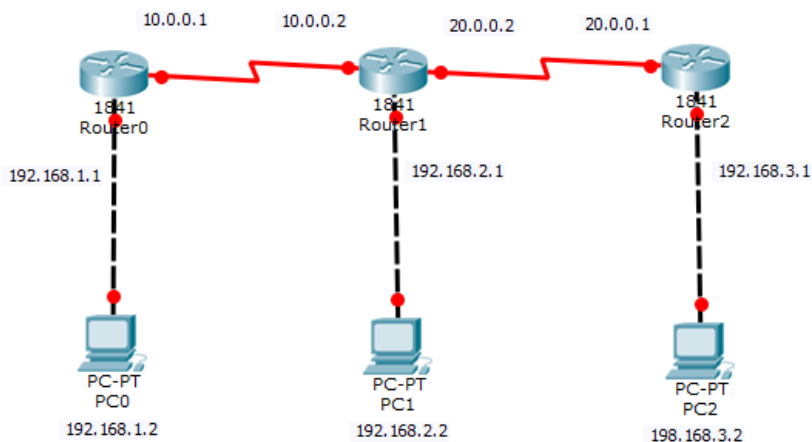
(config)# router ospf<process ID>

(config-router)# network<network ID><wildcard mask>area<area ID>

Implement the OSPF Single Area Network:

Step 1: Initialize the interface and host with IP addresses and default gateway respectively:

- Network topology consists of 3 Host, 3 Interfaces, and 3 switches
- HOST1: IP 192.168.1.2, Default gateway: 192.168.1.1
- HOST2: IP 192.168.2.2, Default gateway: 192.168.2.1
- HOST3: IP 192.168.3.2, Default gateway: 192.168.3.1



As we can see we have configured interface 1 (Router0) with Host 1 which is PC0 and the Serial port.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router0
Router0(config)#interface FastEthernet0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown

Router0(config-if)#exit
Router0(config)#
Router0(config)#interface Serial0/0/0
Router0(config-if)#ip address 10.0.0.1 255.255.255.252
Router0(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Router0(config-if)#exit
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router0(config)#interface FastEthernet0/0
Router0(config-if)#
Router0(config-if)#exit
Router0(config)#interface FastEthernet0/1
Router0(config-if)#
Router0(config-if)#exit
Router0(config)#interface Serial0/0/0
Router0(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
```

Step 2: Configuring the Interface 2 which is router1.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router1
Router1(config)#
Router1(config)#interface FastEthernet0/0
Router1(config-if)#ip address 192.168.2.1 255.255.255.0
Router1(config-if)#no shutdown

Router1(config-if)#exit
Router1(config)#
Router1(config)#interface Serial0/0/0
Router1(config-if)#ip address 10.0.0.2 255.255.255.252
Router1(config-if)#no shutdown

Router1(config-if)#exit
Router1(config)#
Router1(config)#interface Serial0/0/1
Router1(config-if)#ip address 20.0.0.2 255.255.255.252
Router1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
Router1(config-if)#exit
Router1(config)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
```

Step 3: Configuring the Interface 3 which is router2.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router2
Router2(config)#
Router2(config)#interface FastEthernet0/0
Router2(config-if)#ip address 192.168.3.1 255.255.255.0
Router2(config-if)#no shutdown

Router2(config-if)#exit
Router2(config)#
Router2(config)#interface Serial0/0/0
Router2(config-if)#ip address 20.0.0.1 255.255.255.252
Router2(config-if)#no shutdown

Router2(config-if)#exit
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
|
```

Now comes the main part now we have to configure the OSPF implementation:

Step 1: Configure the Router0 and create router OSPF1 and then add network id with wildcard mask.

```
Router0(config)#router ospf 1
Router0(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router0(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router0(config-router)#exit
Router0(config)#
```

- Verifying the OSPF routing

OSPF shares routing information only with neighbors. We use the show IP **ospf** neighbor command to verify OSPF neighbors. The following image shows the output of this command on R1.

The output includes the following fields.

Neighbor ID

This field displays the RID of the neighbor.

State

This field displays the convergency state. An OSPF router goes through seven states to reach convergency. The Full state in this field verifies the router has reached convergency with the router listed in the neighbour ID field.

Interface

This field displays the local interface connected to the neighbor.

Address

This field displays the IP address of the neighbor.

Dead time

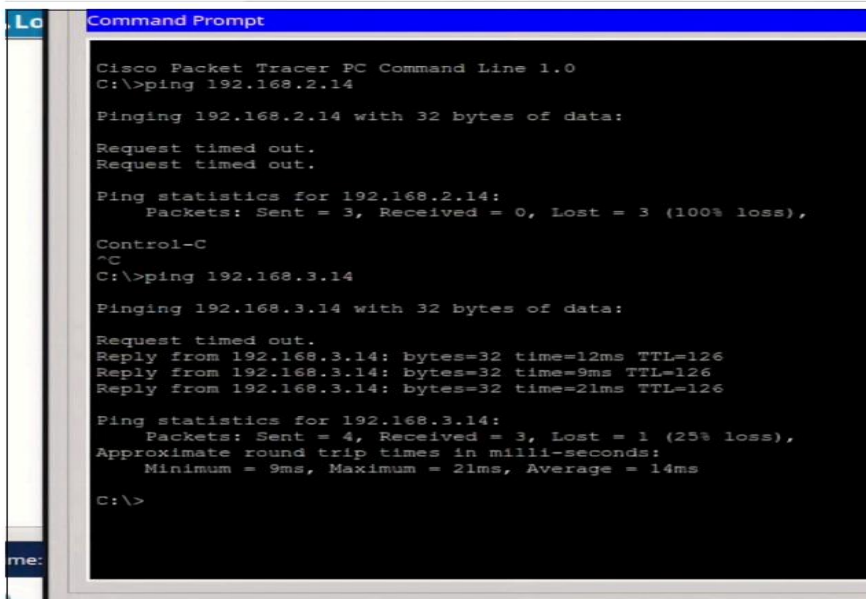
This field displays the dead interval.

- Viewing OSPF routes

The show IP route **ospf** command displays a list of all OSPF routes in the routing table.

- Testing connectivity between end devices

We can test connectivity between end devices to verify the OSPF configuration on all routers. Send ping requests from PC to Server. If it gets replies, it verifies the OSPF configuration. We can also use the **tracert** command to print the path the data packets take to reach the destination.



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.14

Pinging 192.168.2.14 with 32 bytes of data:

Request timed out.
Request timed out.

Ping statistics for 192.168.2.14:
    Packets: Sent = 3, Received = 0, Lost = 3 (100% loss),

Control-C
^C
C:\>ping 192.168.3.14

Pinging 192.168.3.14 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.14: bytes=32 time=12ms TTL=126
Reply from 192.168.3.14: bytes=32 time=9ms TTL=126
Reply from 192.168.3.14: bytes=32 time=21ms TTL=126

Ping statistics for 192.168.3.14:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 21ms, Average = 14ms

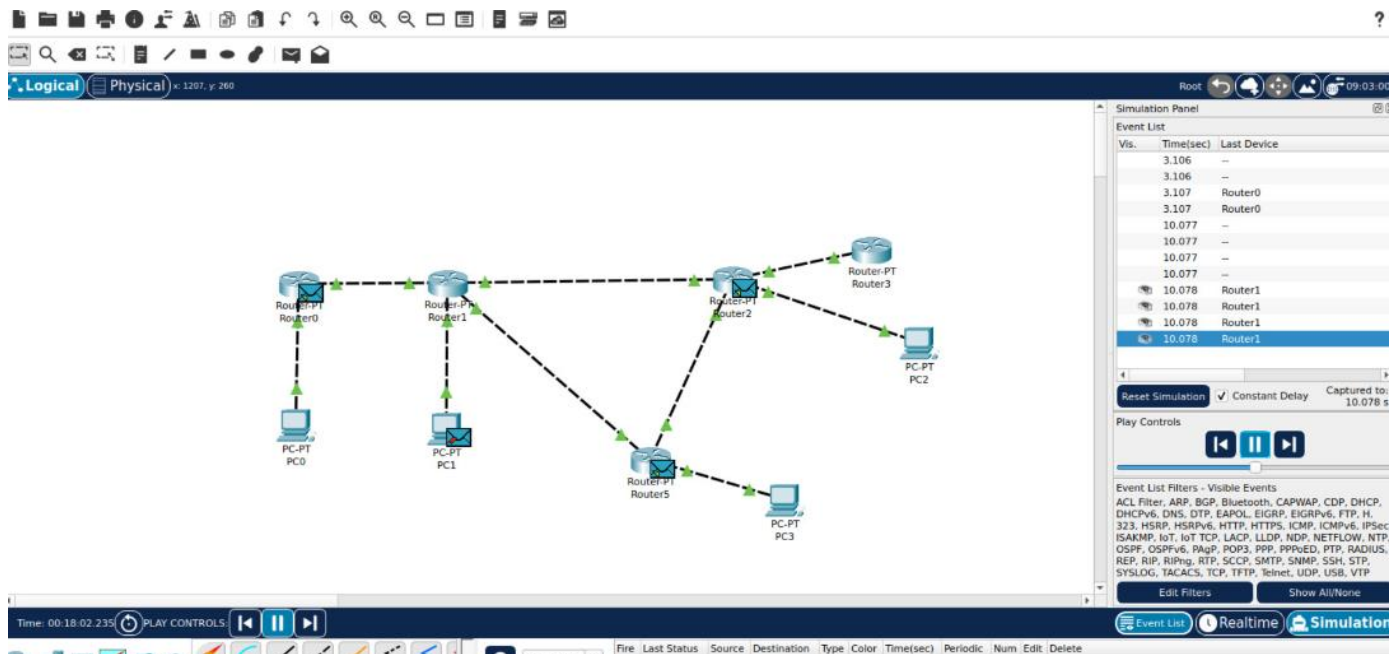
C:\>
  
```

10.c

Configure BGP using packet tracer

BGP (Border Gateway Protocol) is the core routing protocol of the Internet. It is described as a path vector protocol as BGP does not use traditional IGP (OSPF, EIGRP, RIP) metrics, but makes routing decisions based on path, network policies and/or rulesets. It maintains a table of IP networks or 'prefixes' which designate network reachability among autonomous systems (AS). Here three AS are there 1, 71 and 79 respectively. Configure accordingly using three routers.

Step 1: Draw BGP Topology Diagram.



Step 2: Assign IP address on each device as mentioned in Diagram.

Router0

```
Router(config)#interface gigabitethernet0/0
Router(config-if)#ip address 192.168.12.1 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface loopback0

Router(config-if)# ip address 1.1.1.1 255.255.255.255
Router(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Loopback0, changed state to up
```

Router1

```
Router(config)#interface gigabitethernet0/0
Router(config-if)# ip address 192.168.12.2 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitethernet0/1
Router(config-if)# ip address 192.168.23.2 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface loopback0

Router(config-if)# ip address 2.2.2.2 255.255.255.255
Router(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
```

Router2

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitethernet0/0
Router(config-if)# ip address 192.168.23.3 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface loopback0

Router(config-if)# ip address 3.3.3.3 255.255.255.255
Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up

%LINK-5-CHANGED: Interface Loopback0, changed state to up

```

BGP configuration Testing and troubleshooting.

